UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 003 · Types of electrodes · compiled 17 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · Types of electrodes: the map
A cell needs two electrodes; the type depends on the application. Six types: metal–metal ion, metal–insoluble salt–anion, gas, amalgam, redox (quinhydrone a special case), ion-selective. For each, write the reduction reaction and $E = E^\circ - \frac{0.0591}{n}\log\frac{[\text{products}]}{[\text{reactants}]}$ (solids 1, gases as p).

2 · Metal–metal ion electrode
Metal in contact with a solution of its own ions ($\mathrm{Zn|Zn^{2+}}$, $\mathrm{Cu|Cu^{2+}}$, $\mathrm{Ag|Ag^{+}}$). $\mathrm{M^{n+} + ne^- \rightleftharpoons M}$; $E = E^\circ + \frac{2.303RT}{nF}\log[\mathrm{M^{n+}}]$.

3 · Metal–metal insoluble salt–anion electrode
Metal + sparingly soluble salt of the metal, in a solution of a soluble salt of the same anion. Calomel $\mathrm{Hg|Hg_2Cl_2|KCl}$; $\mathrm{Ag|AgCl(s)|HCl}$: $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$, $E = E^\circ - 0.0591\log[\mathrm{Cl^-}]$. Reversible to the anion; used as reference electrodes.

4 · Gas electrode
Gas bubbled over inert Pt foil in a solution of the ions to which the gas is reversible; Pt gives contact and helps set up the gas–ion equilibrium. $\mathrm{Pt|H_2|H^+}$: $\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$, $E = E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[H^+]^2}$; SHE (1 atm, 1 M) $E = 0$. Also $\mathrm{Pt|Cl_2|Cl^-}$.

5 · Amalgam electrode
Like the metal–metal ion electrode but with the metal as an amalgam: $\mathrm{Pb-Hg|Pb^{2+}}$, $E = E^\circ - \frac{0.0591}{2}\log\frac{[\text{Pb-Hg}]}{[\mathrm{Pb^{2+}}]}$. Amalgamation tames reactive metals (Na) and lets the activity be controlled.

6 · Oxidation–reduction electrodes and the quinhydrone electrode

Inert Pt in a solution of both forms of a redox couple ($\mathrm{Fe^{3+}/Fe^{2+}}$, $\mathrm{Ce^{4+}/Ce^{3+}}$, $\mathrm{Sn^{4+}/Sn^{2+}}$); Pt is only a contact; potential from the tendency of one form to change into the other: $E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$. Quinhydrone $\mathrm{Pt|Q,QH_2}$: $E^\circ = 0.699$ V, $E = 0.699 - 0.0591\,\mathrm{pH}$.

7 · Ion-selective (membrane) electrode

An ISE has a membrane that exchanges ions with the solution, so its potential depends on one ion's concentration in the presence of others. Glass electrode (pH meters): $E_G = E_G^\circ + 0.0591\log[\mathrm{H^+}] = E_G^\circ - 0.0591\,\mathrm{pH}$. With SCE: $\mathrm{Hg|Hg_2Cl_2|KCl(sat)||H^+|glass|0.1\,M\,HCl|AgCl|Ag}$.

2 Keyword index

25 terms; the number in brackets is the topic.

Amalgam (5). Alloy of a metal with mercury.

Amalgam electrode (5). $\mathrm{Pb-Hg|Pb^{2+}}$; metal–ion electrode with an amalgam.

Calomel electrode (3). $\mathrm{Hg|Hg_2Cl_2|KCl}$; metal–insoluble salt type; reference.

Chlorine electrode (4). $\mathrm{Pt|Cl_2|Cl^-}$; gas electrode.

Electrical contact (6). The only role of Pt in redox electrodes.

Gas electrode (4). Gas bubbled over inert Pt in a solution of its ions.

Glass electrode (7). ISE selective to $\mathrm{H^+}$; $E_G = E^\circ_G - 0.0591\,\mathrm{pH}$.

Hydrogen electrode (4). $\mathrm{Pt|H_2|H^+}$; $\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$.

Inert metal (4). Pt: conducts, does not react.

Ion-selective electrode (7). Membrane exchanges one ion; potential depends on that ion.

Membrane electrode (7). Another name for an ion-selective electrode.

Metal–insoluble salt–anion electrode (3). Metal + sparingly soluble salt, in a solution of the same anion.

Metal–metal ion electrode (2). Metal in a solution of its own ions.

Nernst equation (1). $E = E^\circ - \frac{0.0591}{n}\log\frac{[\text{products}]}{[\text{reactants}]}$.

Partial pressure (4). How a gas enters the Nernst equation.

Platinised platinum (4). Pt coated with platinum black; large surface.

Porous junction (3). Contact between an electrode's solution and the outer solution.

Quinhydrone electrode (6). $\mathrm{Pt|Q, QH_2}$; $E^\circ = 0.699$ V; pH dependent.

Redox electrode (6). Pt in both oxidised and reduced forms of a couple.

Reduction reaction (1). Always the form in which an electrode reaction is written.

Saturated calomel electrode (SCE) (7). Calomel in saturated KCl, 0.2422 V.

Silver–silver chloride electrode (3). $\mathrm{Ag|AgCl(s)|Cl^-}$; $E = E^\circ - 0.0591\log[\mathrm{Cl^-}]$.

Sparingly soluble salt (3). AgCl, $\mathrm{Hg_2Cl_2}$: the salt layer on the metal.

Standard hydrogen electrode (4). 1 atm $\mathrm{H_2}$, 1 M $\mathrm{H^+}$; $E^\circ = 0$.

Stannic–stannous electrode (6). $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$.

3 Rapid recall

1	How many electrodes make a cell?	Two (two half-cells).
2	Six types of electrodes?	Metal–metal ion, metal–insoluble salt–anion, gas, amalgam, redox, ion-selective.
3	In what form is an electrode reaction written?	As a reduction.
4	Ag/AgCl reaction?	$\mathrm{AgCl} + \mathrm{e}^- \rightleftharpoons \mathrm{Ag} + \mathrm{Cl}^-$.
5	Ag/AgCl Nernst equation?	$E = E^\circ - 0.0591 \log[\mathrm{Cl}^-]$.
6	Example of a metal–insoluble salt electrode besides Ag/AgCl?	Calomel, $\mathrm{Hg}\vert\mathrm{Hg_2Cl_2}\vert\mathrm{KCl}$.
7	Role of Pt in a gas electrode?	Electrical contact; helps set up the gas–ion equilibrium.
8	Hydrogen electrode Nernst equation?	$E = E^\circ - \frac{0.0591}{2} \log \frac{p_{H_2}}{[\mathrm{H}^+]^2}$.
9	SHE conditions?	1 atm $\mathrm{H_2}$, 1 M H^+, $E = 0$.
10	$\mathrm{H_2}$ electrode at pH 3, 2 atm?	−0.1862 V.
11	Why use an amalgam?	To lower and control the activity of a reactive metal (e.g. Na).
12	Lead amalgam representation?	$\mathrm{Pb}-\mathrm{Hg}\vert\mathrm{Pb}^{2+}$.
13	Redox electrode examples?	$\mathrm{Pt}\vert\mathrm{Fe}^{3+},\mathrm{Fe}^{2+}$; $\mathrm{Pt}\vert\mathrm{Ce}^{4+},\mathrm{Ce}^{3+}$; $\mathrm{Pt}\vert\mathrm{Sn}^{4+},\mathrm{Sn}^{2+}$.
14	Stannic–stannous Nernst equation?	$E = E^\circ - \frac{0.0591}{2} \log \frac{[\mathrm{Sn}^{2+}]}{[\mathrm{Sn}^{4+}]}$.
15	Quinhydrone standard potential?	+0.699 V.
16	Quinhydrone potential at pH 4?	0.4626 V.
17	What is an ion-selective electrode?	A membrane electrode whose membrane exchanges one ion, so its potential depends on that ion.
18	Glass electrode equation?	$E_G = E_G^\circ + 0.0591 \log[\mathrm{H}^+] = E_G^\circ - 0.0591\,\mathrm{pH}$.
19	Glass + SCE cell (PESU form)?	$\mathrm{Hg}\vert\mathrm{Hg_2Cl_2}\vert\mathrm{KCl(sat)}\,\Vert\,\mathrm{H}^+\vert\mathrm{glass}\vert 0.1\,\mathrm{M\,HCl}\vert\mathrm{AgCl}\vert\mathrm{Ag}$.
20	Ag/AgCl at 0.1 M chloride?	0.2811 V (with E° = 0.222 V).

Mnemonic (study aid). *"Metal reacts, Pt watches, membrane swaps."* The three groups: metal–ion, amalgam and insoluble-salt electrodes; gas and redox electrodes on inert Pt; ion-selective membranes (a study aid).

4 Last 10 minutes before the exam

Top facts

1. Six types: metal–metal ion, metal–insoluble salt–anion, gas, amalgam, redox (quinhydrone), ion-selective.
2. Always write the reduction; $E = E^\circ - \frac{0.0591}{n}\log\frac{\text{products}}{\text{reactants}}$; solids = 1, gases as p.
3. $\mathrm{Ag/AgCl}$: $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$ (Ag, not $\mathrm{Ag^+}$); $E = E^\circ - 0.0591\log[\mathrm{Cl^-}]$.
4. Gas: Pt + gas + its ions; $E = E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2}$; SHE = 0 V.
5. Amalgam keeps [Pb-Hg] in the log term; amalgamation tames reactive Na.
6. Redox: Pt in both forms; $E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$.
7. Quinhydrone: $E^\circ = 0.699$ V; $E = 0.699 - 0.0591\,\mathrm{pH}$ (below pH 8).
8. ISE: membrane exchanges one ion. Glass: $E_G = E_G^\circ - 0.0591\,\mathrm{pH}$.
9. Glass + SCE: $\mathrm{Hg|Hg_2Cl_2|KCl(sat)||H^+|glass|0.1\,M\,HCl|AgCl|Ag}$; $E_{cell} = E_G - E_{SCE}$.
10. ESA: notes on redox + gas electrodes (Dec 2023); ISE + glass|SCE cell (Jun 2024, Jan 2025).

Top formulas

E1.1 · Nernst equation for any electrode (reduction form) ESA ★★

$$E = E^\circ - \frac{2.303RT}{nF}\log\frac{[\text{products}]}{[\text{reactants}]} = E^\circ - \frac{0.0591}{n}\log\frac{[\text{products}]}{[\text{reactants}]}\ \ (298\ \mathrm{K})$$

GIVES	The single-electrode (reduction) potential at any concentration.
APPLIES	Write the electrode reaction as a reduction first; pure solids and pure metals count as 1, gases enter as partial pressure in atm.
UNITS	V
NOTE	$2.303RT/F = 0.0591$ V at 298 K (0.05914 with $R = 8.314$, $F = 96485$).

T003 00:01:38 · T003 00:07:50

E3.1 · Silver–silver chloride electrode ESA ★

$$E_{\mathrm{Ag/AgCl/Cl^-}} = E^\circ_{\mathrm{Ag/AgCl/Cl^-}} - \frac{0.0591}{1}\log[\mathrm{Cl^-}]$$

GIVES	Potential of $\mathrm{Ag|AgCl(s)|Cl^-}$.
APPLIES	$\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$; $n = 1$.
UNITS	V
NOTE	Reversible to $\mathrm{Cl^-}$: E falls as $[\mathrm{Cl^-}]$ rises. $E^\circ \approx 0.222$ V (given data).

S003·4 · T003 00:04:57

E4.1 · Hydrogen electrode

ESA ★★

$$E_{\mathrm{Pt/H_2/H^+}} = E^\circ_{\mathrm{Pt/H_2/H^+}} - \frac{0.0591}{2}\log\frac{p_{\mathrm{H_2}}}{[\mathrm{H^+}]^2}$$

GIVES Potential of $\mathrm{Pt|H_2(p)|H^+}$.

APPLIES $2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{H_2}$; $n = 2$; $E^\circ = 0$ by convention.

UNITS V

NOTE SHE: $p_{H_2} = 1$ atm, $[\mathrm{H^+}] = 1$ M, so the log term is 0 and $E = E^\circ = 0$.

S003·5 · T003 00:07:50 · T003 00:08:38

E6.1 · Stannic–stannous redox electrode

ESA ★★

$$E_{\mathrm{Pt/Sn^{4+}/Sn^{2+}}} = E^\circ_{\mathrm{Pt/Sn^{4+}/Sn^{2+}}} - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$$

GIVES Potential of $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$.

APPLIES $\mathrm{Sn^{4+}} + 2\mathrm{e^-} \rightleftharpoons \mathrm{Sn^{2+}}$; $n = 2$.

UNITS V

NOTE Only the ratio [reduced]/[oxidised] matters. $E^\circ = 0.15$ V (given data).

S003·7 · T003 00:13:31

E6.3 · Quinhydrone electrode

ESA ★★

$$E_{\mathrm{Pt/Q/QH_2}} = E^\circ_{\mathrm{Pt/Q/QH_2}} - \frac{0.0591}{2}\log\frac{[\mathrm{QH_2}]}{[\mathrm{Q}][\mathrm{H^+}]^2}, \qquad E^\circ_{\mathrm{Pt/Q/QH_2}} = +0.699\ \mathrm{V}$$

GIVES Potential of $\mathrm{Pt|Q, QH_2}$.

APPLIES $\mathrm{Q} + 2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{QH_2}$; $n = 2$.

UNITS V

NOTE $\mathrm{H^+}$ is a reactant, so it sits in the denominator, squared.

S003·8 · T003 00:15:20

E7.1 · Glass electrode (pH-sensitive ISE)

ESA ★★

$$E_G = E^\circ_G + 0.0591\log_{10}[\mathrm{H^+}] = E^\circ_G - 0.0591\,\mathrm{pH}$$

GIVES Potential of the glass electrode.

APPLIES Glass membrane selective to $\mathrm{H^+}$; 298 K.

UNITS V

NOTE E°_G differs for each electrode and is found by calibration with a buffer (Unit 3).

S003·9 · T003 00:17:47 · PDF p.56